

Inverter (LT3462A): Working Document

Digital Audio Player • Rev 6

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1. Summary

Field	Value
Manufacturer / MPN	Analog Devices, LT3462AES6
Package	6-lead TSOT-23 (S6)
Function	2.7 MHz current-mode inverting converter with integrated Schottky; +3V9 → -3.9 V (LT3093 input)

2. Design Requirements

Requirement	Value
Input	+3V9 from LTC3440, 3.827 to 3.981 V (minimum 3.74 V during a load step)
Output	-3.9 V, regulated
Load	LT3093 only
Output current	≈ 27 mA quiescent; ≈ 98 mA peak (0.6 V RMS maximum); 125 mA LT3093 limit
Topology	Dual-inductor inverting (Ćuk)
Switching frequency	2.7 MHz fixed (2.0 to 3.5 MHz), no Burst Mode
Enable	Follows +3V9 (SDREF internal pull-up); no separate shutdown control
Constraints	Low noise · Small footprint

3. Pin Configuration

Pin	Name	Connection
1	SW	L1, C1 (+)
2	GND	GND
3	FB	R1 / R2 midpoint, C4 to -3V9
4	SDREF	R2 to FB only; no GPIO (absolute maximum 2.5 V), no other load
5	D	C1 (-), L2
6	VIN	+3V9, C _{IN}

4. External Components

Ref	Value / Rating	MPN	AL-PN
L1, L2	22 μ H $\pm$ 20%, shielded, 300 mA rated, 400 mA I_{sat} , 810 m Ω max, 1210	Murata LQW32FT220M2HL	#1-090
C1 (flying)	1 μ F $\pm$ 10% 35 V X5R 0402	Murata GRM155R6YA105KE11D	#1-092
C _{IN}	10 μ F 10 V X7R 0805 (6.76 μ F effective at 3.9 V)	Murata GCM21BR71A106KE22K	#1-081
C _{OUT}	10 μ F 10 V X7R 0805 (6.76 μ F effective at 3.9 V)	Murata GCM21BR71A106KE22K	#1-081
R1	309 k Ω 0.1% 1/16 W 0402	YAGEO RE0402BRE07309KL	#1-091
R2	100 k Ω 0.1% 0603	YAGEO RT0603BRD07100KL	#1-066
C4	22 pF $\pm$ 5% 50 V COG 0402 (confirmed by load-step simulation)	Murata GRM1555C1H220JA01D	#1-093

5. Equations and Results

Output voltage (R1 from FB to V_{OUT}, R2 from SDREF to FB)

$$V_{OUT} = -1.265 \text{ V} \cdot \frac{R_1}{R_2} \Rightarrow R_1 = 100 \text{ k}\Omega \cdot \frac{3.9}{1.265} = 308.3 \text{ k}\Omega \rightarrow 309 \text{ k}\Omega$$

$$V_{OUT} = -1.265 \times \frac{309}{100} = -3.909 \text{ V} \quad (-3.816 \text{ to } -3.971 \text{ V over SDREF - FB} = 1.235 \text{ to } 1.285 \text{ V})$$

Source: [LT3462/LT3462A Datasheet \(Analog Devices\)](#), output voltage programming.

Divider current (SDREF can source 80 μ A)

$$I_{R2} = \frac{1.265 \text{ V}}{100 \text{ k}\Omega} = 12.7 \mu\text{A}$$

Source: [LT3462/LT3462A Datasheet \(Analog Devices\)](#).

Duty cycle (V_D = 0.8 V Schottky, V_{CE} = 0.27 V switch)

$$D = \frac{|V_{OUT}| + V_D}{V_{IN} - V_{CE} + |V_{OUT}| + V_D} = \frac{4.709}{3.634 + 4.709} = 56.4\% \quad (57.6\% \text{ at } 3.74 \text{ V; max } 77\%)$$

Source: [LT3462/LT3462A Datasheet \(Analog Devices\)](#), duty cycle.

Inductor ripple (each 22 µH inductor)

$$\Delta I_L = \frac{V_{IN} D}{f L} = \frac{3.904 \times 0.564}{2.7 \text{ MHz} \times 22 \mu\text{H}} = 37 \text{ mA}_{pp} \quad (50 \text{ mA}_{pp} \text{ at } 2.0 \text{ MHz})$$

Inductor and switch current (98 mA output, $\eta = 0.65$ assumed)

$$I_{L1} = \frac{|V_{OUT}| I_{OUT}}{\eta V_{IN}} = \frac{3.909 \times 0.098}{0.65 \times 3.904} = 151 \text{ mA}, \quad I_{L2} = I_{OUT} = 98 \text{ mA}$$
$$I_{SW,pk} = I_{L1} + I_{L2} + \Delta I_L = 151 + 98 + 37 = 286 \text{ mA} \quad (299 \text{ mA at } 2.0 \text{ MHz})$$

Source: [LT3462/LT3462A Datasheet \(Analog Devices\)](#), switch current.

Output current	Switch peak vs 300 mA min / 420 mA typ limit
27 mA (quiescent)	≈ 0.10 A
69 mA (110 dB SPL peaks)	≈ 0.21 A
98 mA (full scale, 23.8 Ω)	≈ 0.29 A: at the minimum limit
125 mA (LT3093 limit)	≈ 0.36 A: the inverter limits first

Inrush (undamped upper bound; Schottky surge limit 1.5 A)

$$I_p \leq V_{IN} \sqrt{\frac{C_1}{L_1}} = 3.9 \text{ V} \cdot \sqrt{\frac{1 \mu\text{F}}{22 \mu\text{H}}} = 0.83 \text{ A}$$

Source: step-input bound; the ≈ 6 ms LTC3440 soft-start ramp keeps the real peak well below it.

Inductor copper loss (98 mA output, 810 mΩ max)

$$P_{DCR} = (I_{L1}^2 + I_{L2}^2) R_{DCR} = (0.151^2 + 0.098^2) \times 0.81 = 26 \text{ mW}$$

5.1 Simulation Results

Item (3.9 V in)	98 mA load (39.9 Ω)	27 mA load (144.8 Ω)
V_{OUT}	-3.929 V	-3.938 V
Efficiency	67.7%	71.8%
Input power	572 mW	149 mW
I_{L1} average (LTC3440 load)	141 mA	35 mA
Switch peak	0.282 A	0.100 A
Output ripple	0.52 mV pp	0.78 mV pp

C4	Step up (27 → 98 mA)	Step down (98 → 27 mA)
18 pF	+28 mV (to -3.910 V)	-20 mV (to -3.958 V)
22 pF	+26 mV (to -3.912 V)	-18 mV (to -3.956 V)

Load steps are well damped with a single overshoot. At the LT3093 output this is ≈ 10 µV (68 dB line-step rejection).

6. Important Values

Item	Value
Input range	2.5 to 16 V
Switch current limit	300 mA min / 420 mA typ (varies with duty cycle)
Switch V_{CESAT}	270 mV typ / 350 mV max at 250 mA
Schottky forward drop	800 mV typ / 1100 mV max at 250 mA
Maximum duty cycle	77%
SDREF reference	1.265 V (1.245 to 1.285 V), up to 80 μ A
Supply current	2.9 mA typ not switching; < 10 μ A shutdown
Load regulation	-3.938 V at 27 mA $\rightarrow$ -3.929 V at 98 mA (simulated)
Switch peak at 98 mA	0.282 A simulated vs 300 mA min (6% margin)
IC dissipation at 98 mA	$\approx$ 160 mW $\rightarrow \approx$ 24 $^{\circ}$ C rise (θ_{JA} 150 $^{\circ}$ C/W)

Source: [LT3462/LT3462A Datasheet \(Analog Devices\)](#).

7. Open Problems

None.

8. Revision Log

Rev	Date	Change
0	2026-09-23	Initial issue: part selection, divider, duty cycle, switch current, inrush.
1	2026-09-23	L1/L2 LQW32FT220M2HL; SDREF to R2 only.
2	2026-09-23	L1/L2 library part; I_{sat} 400 mA confirmed.
3	2026-09-23	C1 1 μ F $\geq$ 25 V; efficiency and current simulation.
4	2026-09-23	Load-step simulation; C4 = 22 pF confirmed.
5	2026-09-23	Library parts assigned: R1, C1, C4.
6	2026-09-25	Reformatted; linked contents and sources; open problems closed.

9. Sources

1. [LT3462/LT3462A Datasheet \(Analog Devices\)](#)